Indian Ocean SST and Dipole Outlook

Forecast issued 31 August 2026 · covering 1–30 September 2026

ACCORD / CHC — draft for review

8 September 2026

Abstract

This is a forecast of sea surface temperature in the two Indian Ocean Dipole boxes, and of the dipole itself, for the next 30 days and for each of the next four weeks. It uses the ECMWF sub-seasonal (S2S) forecast, corrected against NOAA OI SST by linear regression.

The forecast is for a positive Indian Ocean Dipole at every horizon. The dipole index is +0.42 °C in week 1, rising to about +0.84 °C in weeks 2 and 3, and +0.73 °C averaged over the next 30 days. This is driven almost entirely by a warm western box: western Indian Ocean temperature is forecast at 28.4–28.8 °C, which is 0.7–1.0 °C above its 2006–2025 average for these dates. The eastern box is close to normal, 0.1–0.3 °C above average.

The forecast is more accurate than persistence at every horizon, and the margin grows with lead time. Tested on twenty years of past forecasts, the model's correlation with what actually happened is 0.72–0.91, against 0.31–0.85 for simply assuming the ocean stays as it is now. In the eastern box at weeks 3 and 4, persistence falls to about 0.31 while the model holds 0.77–0.82. At week 1 the model is only slightly better than persistence, so the useful range of this product is weeks 2 to 4.

Method

Forecast model. ECMWF S2S extended-range ensemble, issued 31 August 2026: 100 members, 46 daily steps, 1.5° grid. From the ECMWF Data Store (`ecds.ecmwf.int`), dataset `s2s-forecasts`, variable `sea_surface_temperature`.

Training data. The reforecast set for the same calendar date (`s2s-reforecasts`) — the same run for 31 August in each of the twenty years 2006–2025, 10 members each. The correction is fitted on these twenty past forecasts.

Observations. NOAA OI SST v2.1, 0.25° daily, from NOAA PSL — the record CHC uses.

Boxes. Saji et al. (1999) dipole boxes. West 50–70 °E, 10 °S–10 °N; east 90–110 °E, 10 °S–0. Averages are area-weighted by cosine of latitude on each dataset's own grid, so no regridding is needed to compare them.

Lead windows. Week 1 is forecast days 1–7, then 8–14, 15–21, 22–28, and 1–30. Day 1 is the first full day after the 00Z issuance.

Correction. For each box and window separately, a straight line fitted by ordinary least squares — observed against forecast temperature across the twenty reforecast years — then applied to this year's forecast. Regression rather than quantile matching, because a straight line continues a warming trend outside the historical range while quantile matching pulls values back inside it.

Uncertainty and testing. The 80% interval is the spread of the regression's own errors over the twenty training years: how wrong this method has been, not how much this week's 100 members disagree. All accuracy figures are leave-one-year-out — the line refitted without each year, then used to predict it — so nothing quoted is measured on data used to fit it.

Reproducing. `iod_pipeline.run("2026-08-31")` in `experiments/IOD-forecasting/`; the date is the only input. Data via `acmadDL`, analysis via `africas2s`.

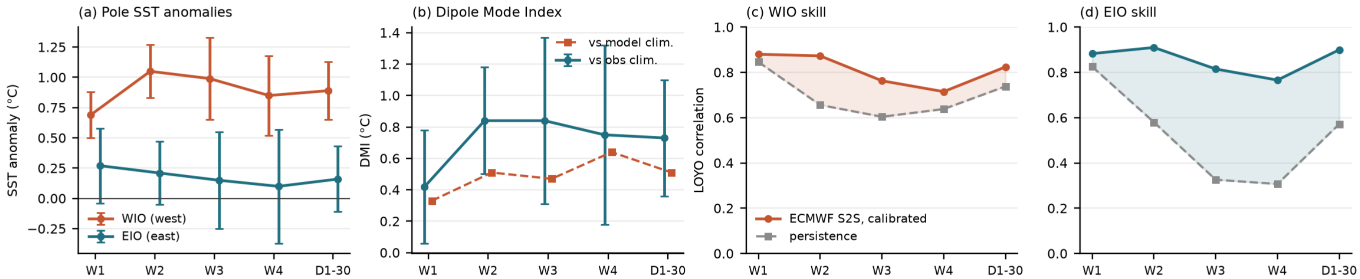

Figure 1: **(a)** Forecast temperature anomaly in each box, with 80% intervals. **(b)** The dipole index against both the observed climatology and the model's own. **(c,d)** Correlation between forecast and observation over twenty past years, leave-one-year-out; grey is persistence — assuming the ocean stays as it is at issue — and the shaded gap is what the model adds.

Results

Horizon	Valid dates	West box	East box	Dipole
Week 1	1–7 Sep	28.35 °C (+0.69)	28.35 °C (+0.27)	+0.42
Week 2	8–14 Sep	28.77 °C (+1.05)	28.32 °C (+0.21)	+0.84
Week 3	15–21 Sep	28.81 °C (+0.99)	28.28 °C (+0.16)	+0.84
Week 4	22–28 Sep	28.81 °C (+0.85)	28.27 °C (+0.10)	+0.75
Days 1–30	1–30 Sep	28.70 °C (+0.89)	28.29 °C (+0.17)	+0.73

Temperatures are absolute; brackets are the departure from the 2006–2025 average for the same dates, dipole in °C. Against the model's own climatology the dipole is smaller (+0.33 to +0.64); the difference between the two is the model's bias at each lead, which the regression removes. Both are shown because neither is more correct — the observed version asks how warm the ocean is against its own recent history, the model version how warm against what this model usually predicts at this range.

Accuracy

	Week 1	Week 2	Week 3	Week 4	Days 1–30
West — model	0.88	0.87	0.76	0.72	0.82
West — persistence	0.85	0.66	0.60	0.64	0.74
East — model	0.88	0.91	0.82	0.77	0.90
East — persistence	0.83	0.58	0.33	0.31	0.57

Typical error is 0.16–0.28 °C (west) and 0.21–0.38 °C (east).

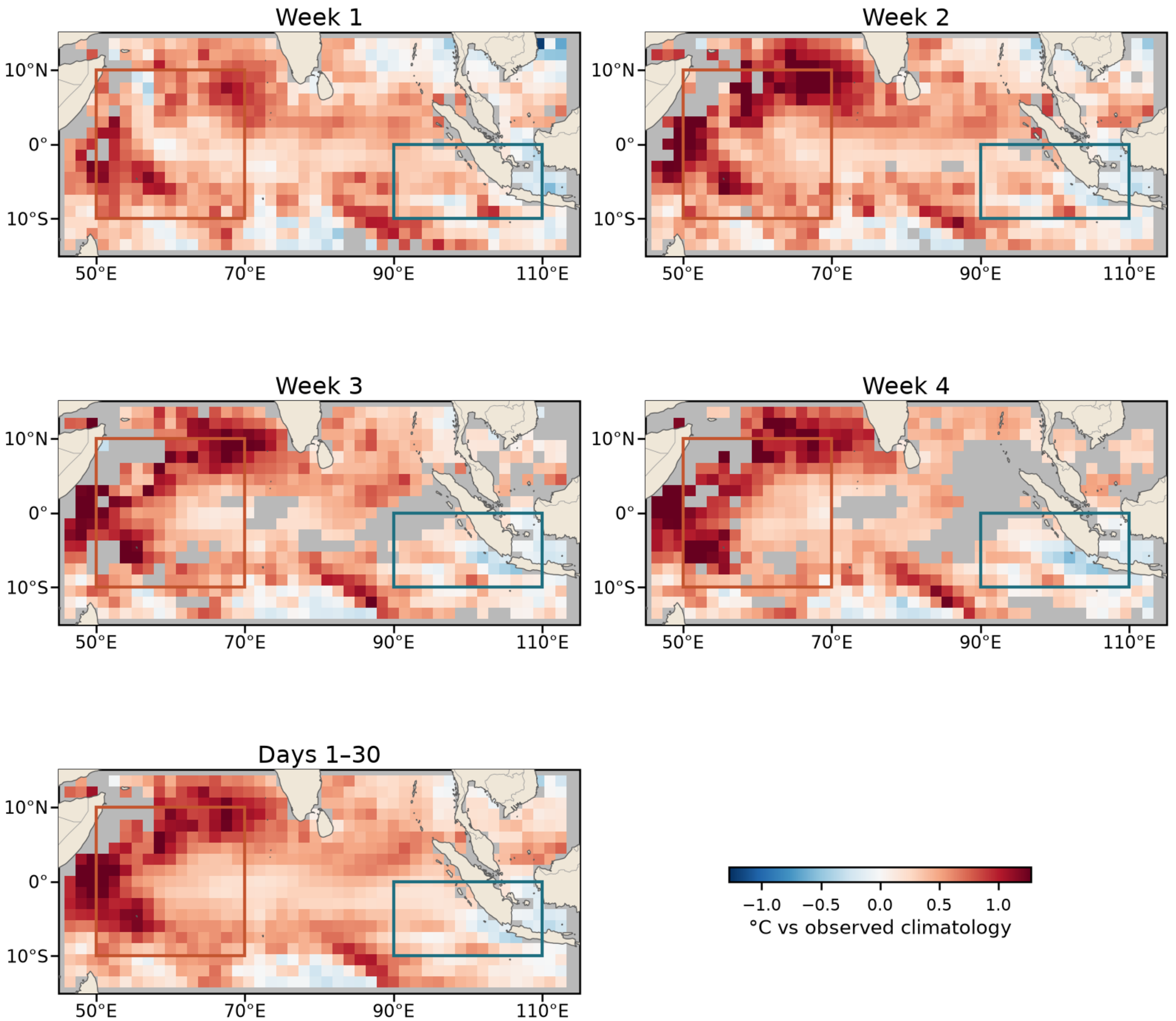

Figure 2: Corrected SST anomaly at every horizon (°C vs the 2006–2025 observed average). Boxes: western (orange) and eastern (teal) poles; land in tan. Grey ocean cells are where leave-one-year-out correlation falls below 0.4 and the forecast should not be relied on — note how that area grows from week 1 to week 4. Each cell is corrected by its own regression; observations are averaged from 0.25° to the model's 1.5° grid.

Limitations

- **Week 1 adds little** — at that range assuming no change is nearly as accurate, so the value of this product is weeks 2 to 4.
- **Twenty years is a short training record**, so one unusual year moves a correction. This follows from how ECMWF produces reforecasts, not from a choice we made.
- **The interval is historical**, not ensemble-based; a version using the 100 members' spread is the obvious next step.
- **Map and box numbers differ by 0.1–0.3 °C** — correcting each cell then averaging is not the same calculation as correcting the box average; quote the table for a box.
- **One model only**, ECMWF S2S. GEFSv12 was considered and set aside.